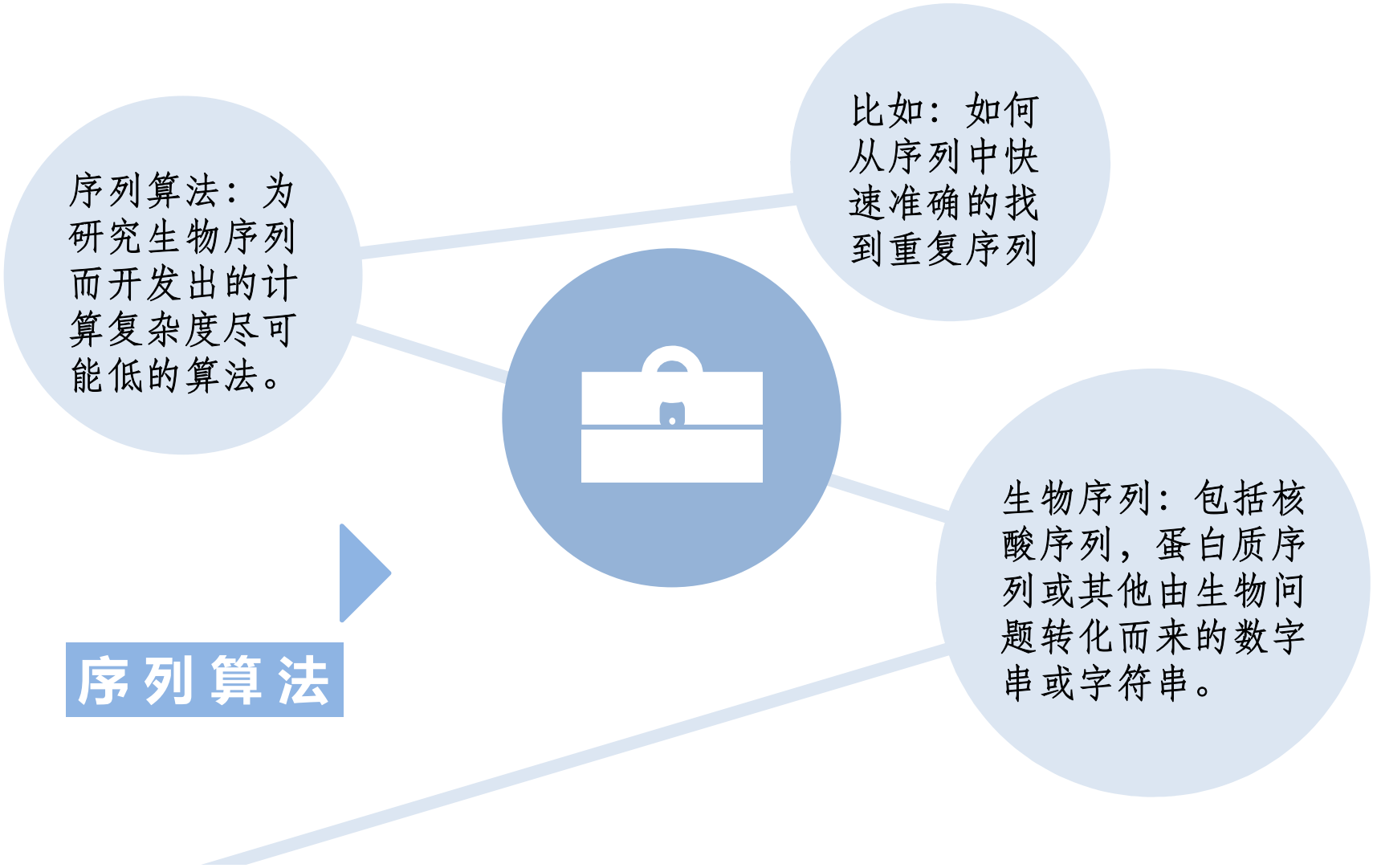

基本序列算法

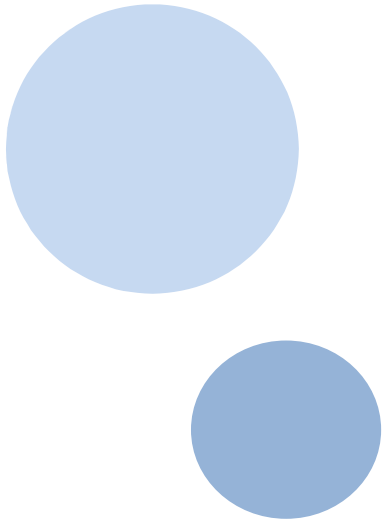

序列算法：为研究生物序列而开发出的计算复杂度尽可能低的算法。
比如：如何从序列中快速准确的找到重复序列
生物序列：包括核酸序列，蛋白质序列或其他由生物问题转化而来的数字串或字符串。
序列算法


构建后缀树

序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S：SDSDFSDFG$
1：SDSDFSDFG$
2：DSDFSDFG$
3：SDFSDFG$
4：DFSDFG$
5：FSDFG$
6：SDFG$
7：DFG$
8：FG$
9：G$
10：$

后缀就是包含最后一个字符的子序列。



序列S： SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S： SDSDFSDFG$

1: SDSDFSDFG$
2: DSDFSDFG$
3: SDFSDFG$
4: DFSDFG$
5: FSDFG$
6: SDFG$
7: DFG$
8: FG$
9: G$
10: $

最后一个字符后面还要加上一个$，表示结尾。

序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S：SDSDFSDFG$

1：SDSDFSDFG$

2：DSDFSDFG$

3：SDFSDFG$

4：DFSDFG$

5：FSDFG$

6：SDFG$

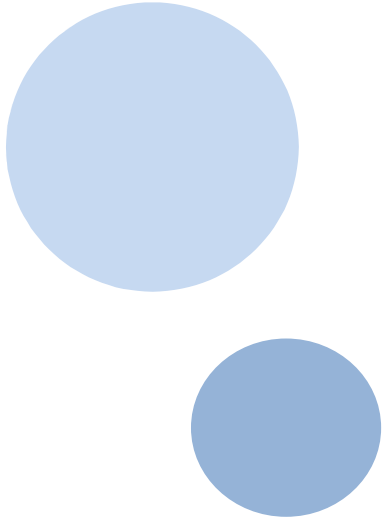
7：DFG$

8：FG$

9：G$

10：$

序列本身就是自己的一个子序列。



序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S：SDSDFSDFG$

1：SDSDFSDFG$

2：DSDFSDFG$

3：SDFSDFG$

4：DFSDFG$

5：FSDFG$

6：SDFG$

7：DFG$

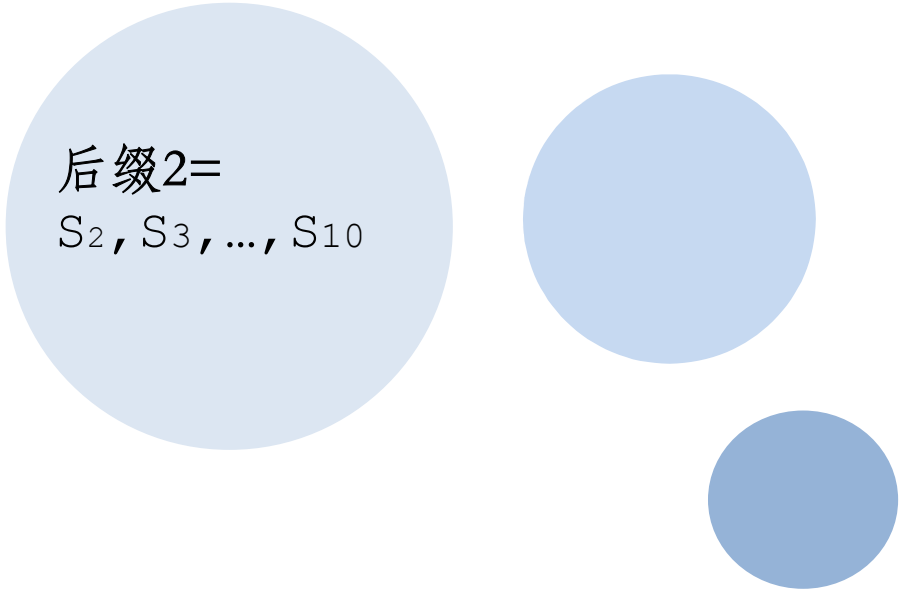
8：FG$

9：G$

10：$

后缀2=

$S_2, S_3, \ldots, S_{10}$



序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S：SDSDFSDFG$

1：SDSDFSDFG$

2：DSDFSDFG$

3：SDFSDFG$

4：DFSDFG$

5：FSDFG$

6：SDFG$

7：DFG$

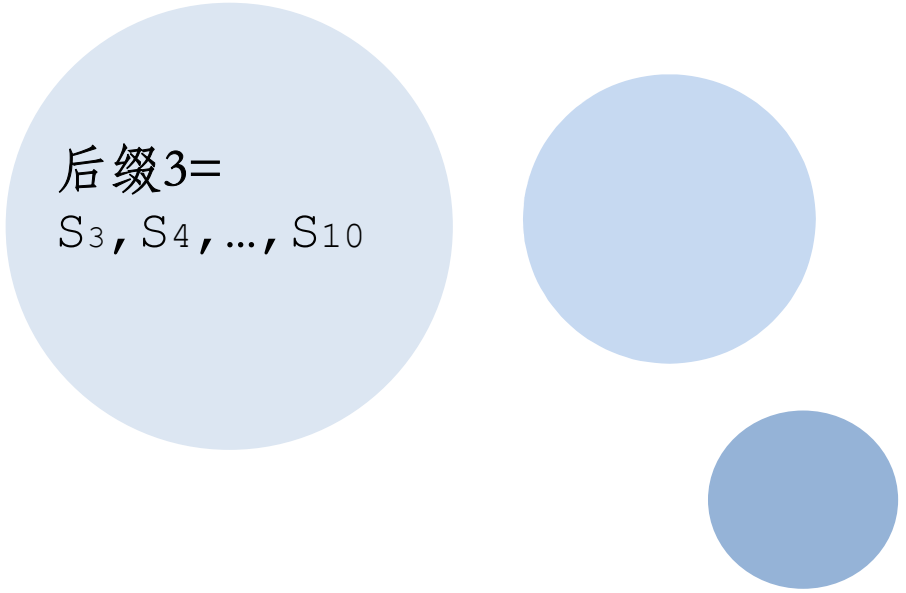
8：FG$

9：G$

10：$

后缀3=

$S_3, S_4, \ldots, S_{10}$

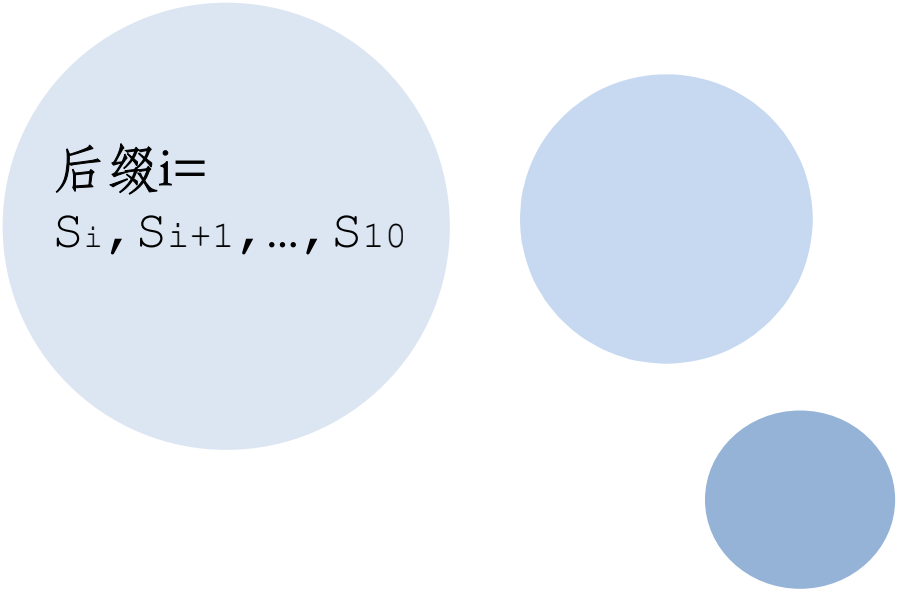


序列S: SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S: SDSDFSDFG$

1: SDSDFSDFG$
2: DSDFSDFG$
3: SDFSDFG$
4: DFSDFG$
5: FSDFG$
6: SDFG$
7: DFG$
8: FG$
9: G$
10: $

后缀i=
$S_i, S_{i+1}, \ldots, S_{10}$



序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S：SDSDFSDFG$

1：SDSDFSDFG$

2：DSDFSDFG$

3：SDFSDFG$

4：DFSDFG$

5：FSDFG$

6：SDFG$

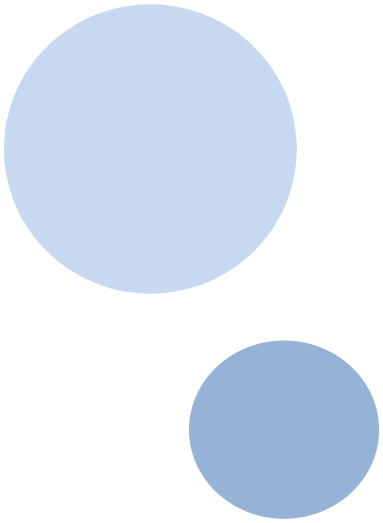
7：DFG$

8：FG$

9：G$

10：$

后缀$是序列S最短的一个后缀。



序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

序列S：SDSDFSDFG$

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5：FSDFG$

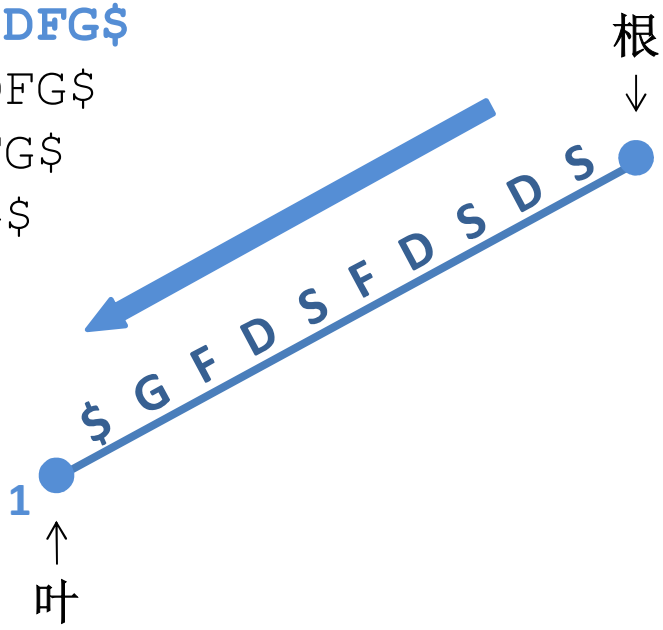
6：SDFG$

7：DFG$

8：FG$

9：G$

10：$



序列S： SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

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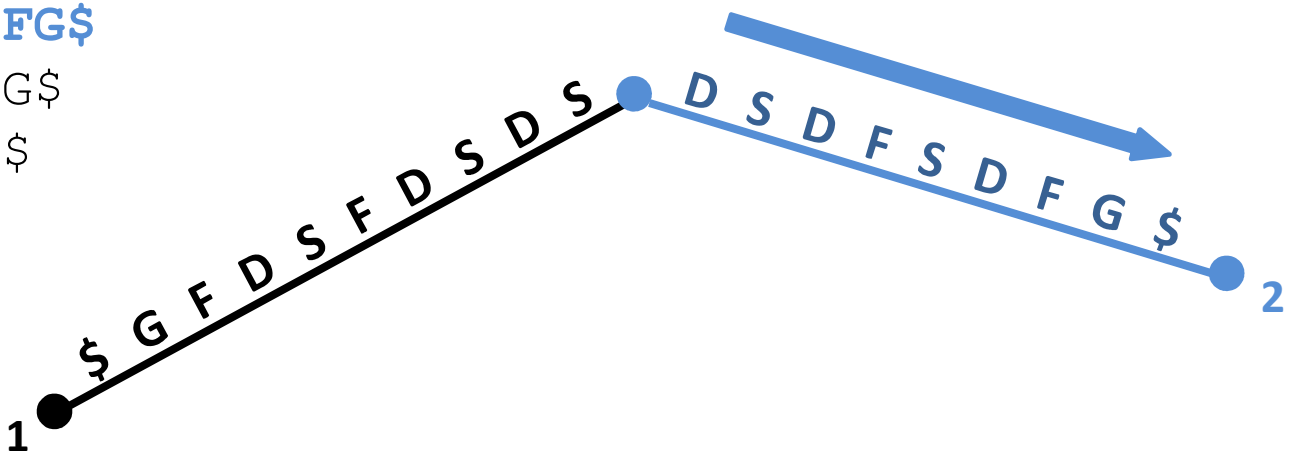
6： SDFG$

7： DFG$

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序列S：SDSDFSDFG　　后缀=$S_{1\le i\le n}, S_{i+1}, S_{i+2}, \ldots, S_n$　　n=length(S)

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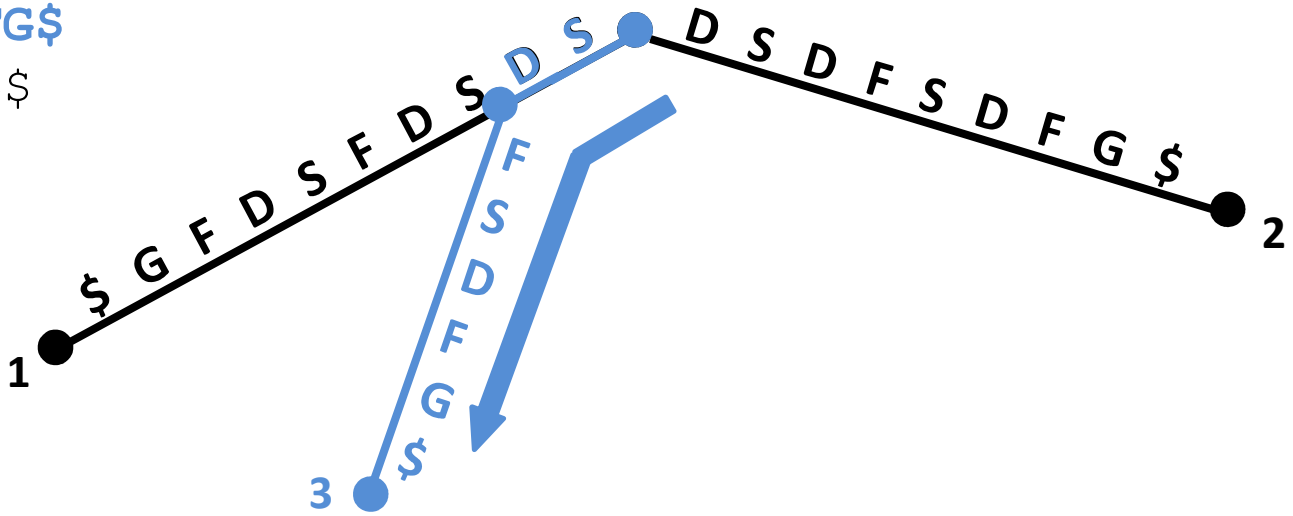
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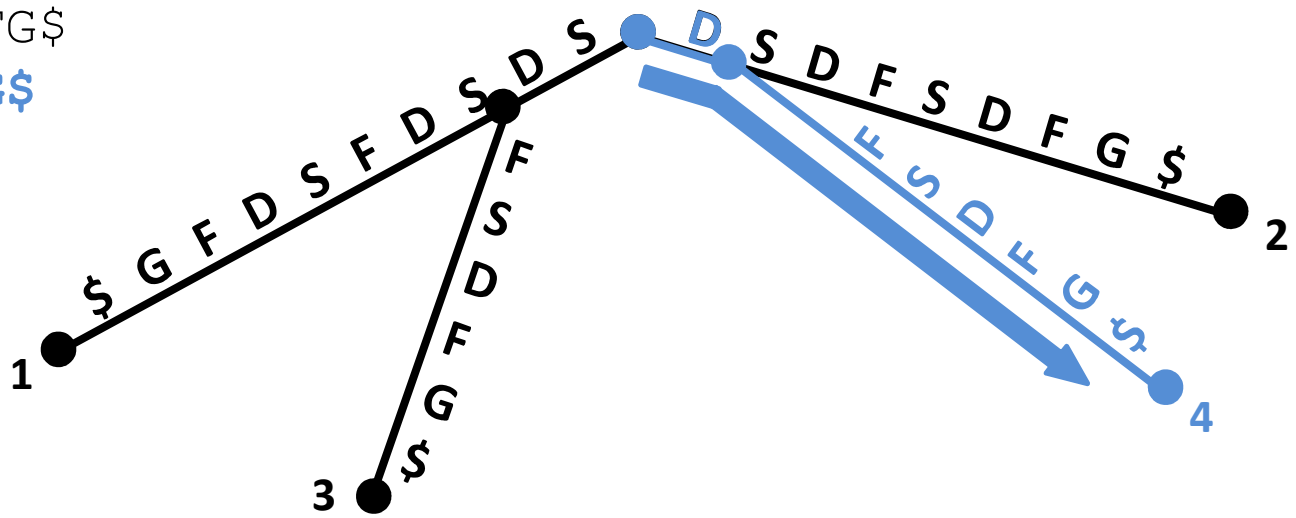
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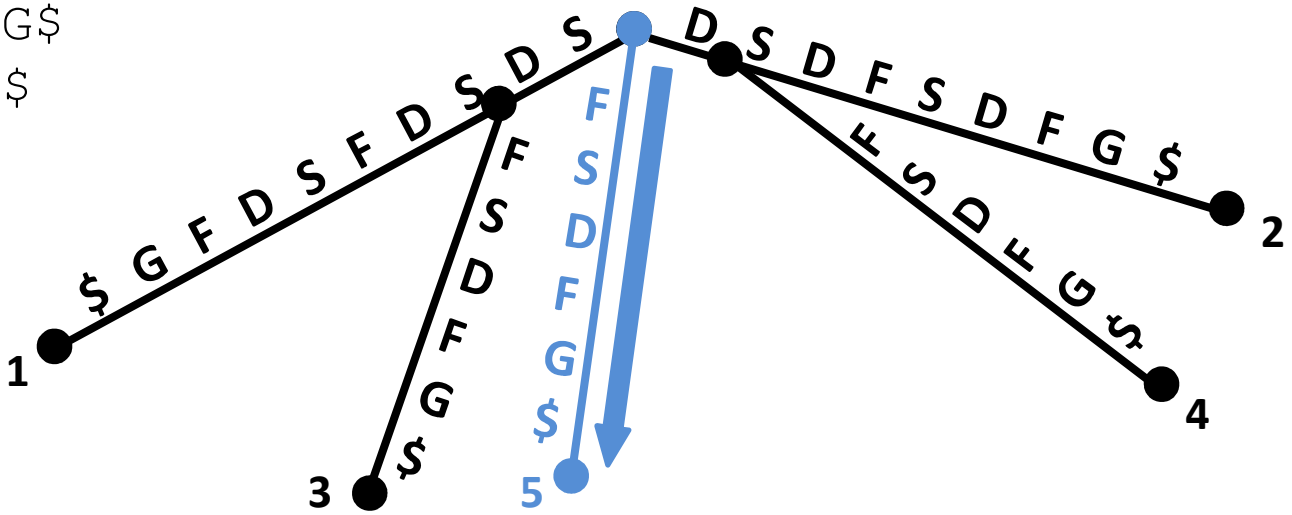
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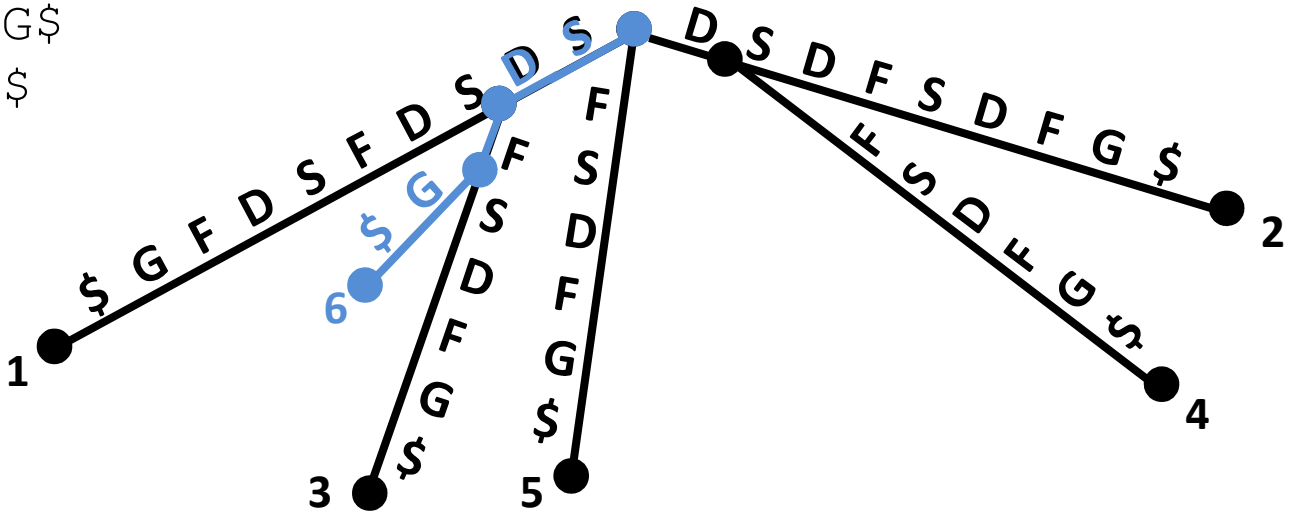
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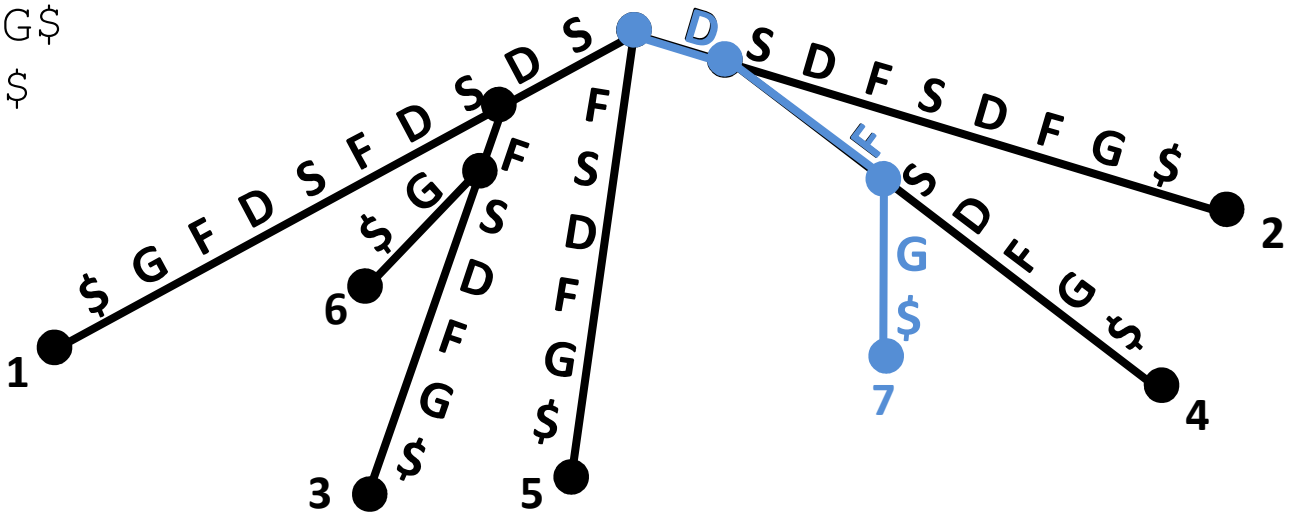
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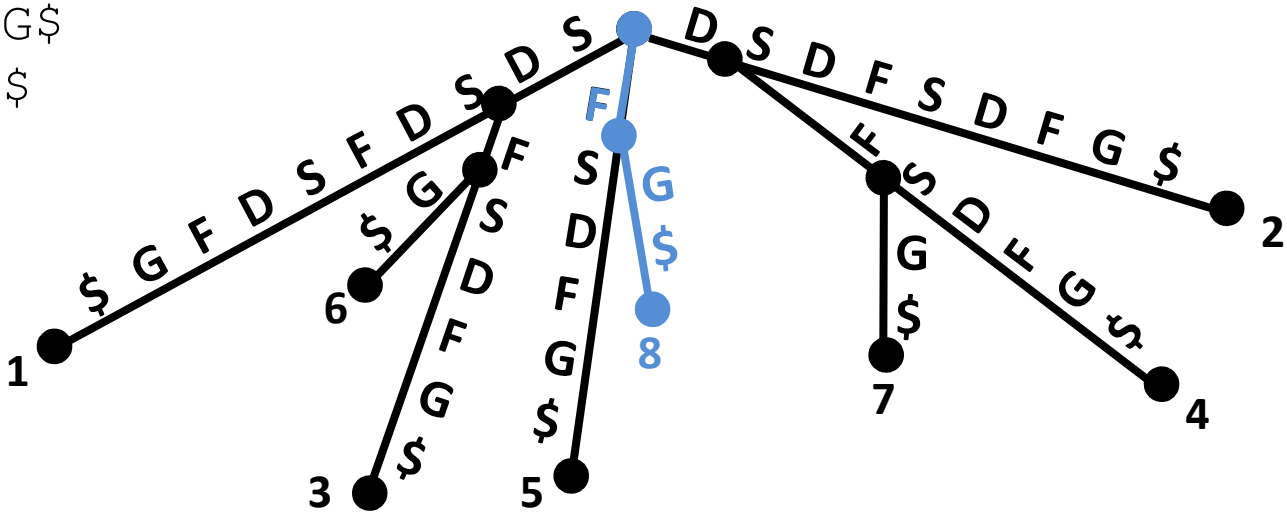
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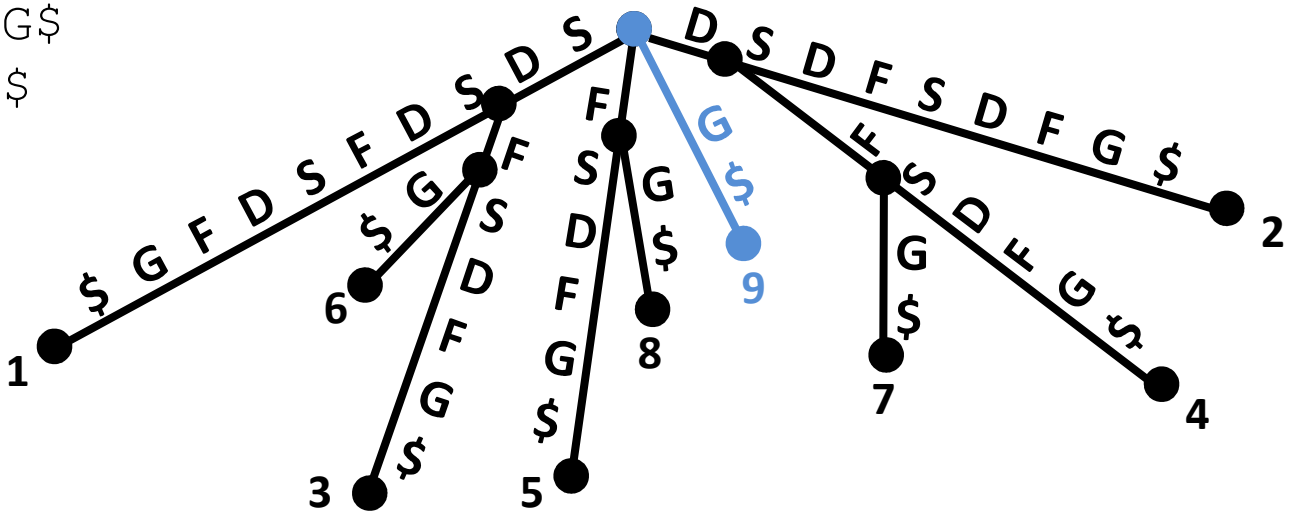
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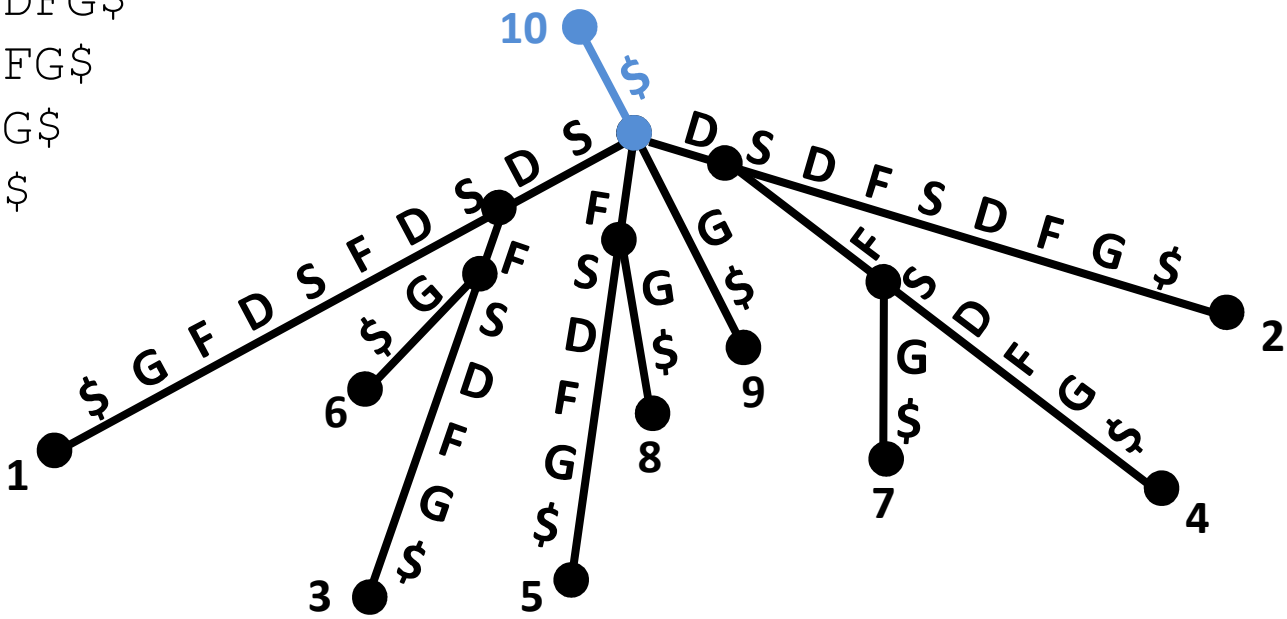
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